Chapter 4 — Work, Energy and Power

Board-style MCQs · PECTAA Physics 11 · Answer key at end

Practice MCQs

1. Work done is zero when force and displacement are:

(a) perpendicular **(b)** opposite **(c)** equal **(d)** parallel

2. Work done by a constant force is:

(a) Fd cos θ **(b)** F + d **(c)** Fd sin θ **(d)** F/d

3. Work is a:

(a) base quantity **(b)** dimensionless quantity **(c)** scalar quantity **(d)** vector quantity

4. SI unit of work is:

(a) pascal **(b)** joule **(c)** newton **(d)** watt

5. Work done by a variable force is found from:

(a) F/d **(b)** F + d **(c)** area under F–d graph **(d)** slope of F–d graph

6. For a spring, work done in stretching is:

(a) kx^2 **(b)** $\frac{1}{2}kx$ **(c)** kx **(d)** $\frac{1}{2}kx^2$

7. A conservative force has zero work over:

(a) a circular path only **(b)** a straight path **(c)** a closed path **(d)** any open path

8. Which is a conservative force?

(a) air resistance **(b)** friction **(c)** gravity **(d)** viscous drag

9. Which is a non-conservative force?

(a) gravity **(b)** friction **(c)** spring force **(d)** electrostatic force

10. Power is:

(a) energy × time **(b)** work × time **(c)** work/time **(d)** force × distance

11. SI unit of power is:

(a) watt **(b)** newton **(c)** pascal **(d)** joule

12. One kilowatt-hour equals:

(a) 3.6×10^3 J **(b)** 1000 J **(c)** 3.6×10^6 J **(d)** 3600 J

13. Kilowatt-hour is a unit of:

(a) power **(b)** force **(c)** time **(d)** energy

14. Power can also be written as:

(a) F·v **(b)** F − v **(c)** F + v **(d)** F/v

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15. Kinetic energy is:

(a) mv **(b)** mv^2 **(c)** mgh **(d)** $\frac{1}{2}mv^2$

16. If speed is doubled, kinetic energy becomes:

(a) two times **(b)** unchanged **(c)** half **(d)** four times

17. If momentum is doubled, kinetic energy becomes:

(a) four times **(b)** unchanged **(c)** two times **(d)** half

18. The work–energy theorem states W_{net} equals:

(a) total energy **(b)** change in kinetic energy **(c)** power **(d)** change in potential energy

19. Gravitational potential energy near Earth is:

(a) mgh **(b)** GMm/r **(c)** −GMm/r **(d)** $\frac{1}{2}mv^2$

20. Absolute gravitational potential energy is:

(a) $\frac{1}{2}mv^2$ **(b)** mgh **(c)** GMm/R **(d)** −GMm/R

21. The negative sign in absolute potential energy shows:

(a) repulsive field **(b)** kinetic energy **(c)** zero energy **(d)** attractive field

22. Escape velocity is:

(a) $\sqrt{(2GM)}$ **(b)** 2gR **(c)** $\sqrt{(2gR)}$ **(d)** $\sqrt{(gR)}$

23. Escape velocity from Earth is about:

(a) 1.0 km s^{-1} **(b)** 3.0 km s^{-1} **(c)** 11.2 km s^{-1} **(d)** 7.9 km s^{-1}

24. Escape velocity is independent of:

(a) mass of Earth **(b)** g **(c)** radius of Earth **(d)** mass of escaping body

25. In the absence of friction, mechanical energy is:

(a) conserved **(b)** increased **(c)** zero **(d)** decreased

26. When friction acts, mechanical energy is:

(a) unchanged **(b)** destroyed **(c)** converted to heat **(d)** created

27. A body at rest may have:

(a) velocity **(b)** kinetic energy **(c)** momentum **(d)** potential energy

28. The work done by gravity in a closed path is:

(a) negative **(b)** 2mgh **(c)** zero **(d)** mgh

29. The area under a force–extension graph gives:

(a) power **(b)** momentum **(c)** stress **(d)** elastic potential energy

30. Elastic potential energy in a spring is:

(a) $\frac{1}{2}kx$ **(b)** $\frac{1}{2}kx^2$ **(c)** kx^2 **(d)** kx

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31. A 6 W lamp in 2 min gives energy:

(a) 20 J **(b)** 720 J **(c)** 12 J **(d)** 360 J

32. A 2 W motor using 3000 J runs for:

(a) 25 min **(b)** 10 min **(c)** 50 min **(d)** 15 min

33. A crane lifts 1000 kg to 100 m in 20 s; power is:

(a) 4.9 kW **(b)** 20 kW **(c)** 49 kW **(d)** 98 kW

34. If work done is 25 J for 5 N force over 10 m, angle is:

(a) 30° **(b)** 45° **(c)** 60° **(d)** 90°

35. The kinetic energy acquired from rest under constant force over fixed distance is:

(a) independent of mass **(b)** zero **(c)** proportional to √m **(d)** proportional to mass

36. A child on a swing from 0.5 m to 1.5 m has max speed about:

(a) 4.4 m s^{-1} **(b)** 9.8 m s^{-1} **(c)** 1.5 m s^{-1} **(d)** depends on mass

37. When a ball is thrown up and falls back, the conservative force is:

(a) friction **(b)** gravity **(c)** hand force **(d)** air resistance

38. According to work–energy principle, work done equals:

(a) zero **(b)** change in P.E. **(c)** sum of K.E. and P.E. **(d)** change in K.E.

39. Absolute and gravitational units of work in MKS are:

(a) erg and joule **(b)** joule and kgf·m **(c)** newton and joule **(d)** watt and joule

40. In CGS, the absolute unit of work is:

(a) erg **(b)** dyne **(c)** watt **(d)** joule

41. A body dropped from H reaches ground with speed 1.2√(gH); work by friction is:

(a) zero **(b)** −mgH **(c)** 0.28 mgH **(d)** −0.28 mgH

42. If bicycle K.E. is 150 J and speed doubles, K.E. becomes:

(a) 150 J **(b)** 300 J **(c)** 75 J **(d)** 600 J

43. If mass triples and speed halves, K.E. becomes:

(a) 0.75 times **(b)** 3 times **(c)** 1.5 times **(d)** 0.5 times

44. Wind contains:

(a) no energy **(b)** kinetic energy **(c)** chemical energy **(d)** potential energy only

45. A slow truck can have more K.E. than a fast car because:

(a) it has larger mass **(b)** it has less momentum **(c)** it has no friction **(d)** it has larger speed

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46. For same momentum, lighter body has:

(a) greater K.E. **(b)** same K.E. **(c)** less K.E. **(d)** zero K.E.

47. A motorcycle moving at constant speed on horizontal track with no net force has:

(a) negative work **(b)** no net work done **(c)** positive work **(d)** maximum work

48. A force reducing speed from 14 to 6 m/s does:

(a) negative work **(b)** zero work **(c)** positive work **(d)** maximum work

49. The international space station has:

(a) both K.E. and P.E. **(b)** only P.E. **(c)** only K.E. **(d)** no energy

50. Work done against friction is non-conservative because:

(a) it is independent of path **(b)** it depends on path **(c)** it is stored as P.E. **(d)** it is zero in closed path

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Answer Key with Brief Reasons

1. Answer: (a) perpendicular — cos 90° = 0.

2. Answer: (a) Fd cos θ — W = F·d.

3. Answer: (c) scalar quantity — Dot product gives scalar.

4. Answer: (b) joule — 1 J = 1 N m.

5. Answer: (c) area under F–d graph — Area gives work.

6. Answer: (d) ½kx² — Average force ½kx times x.

7. Answer: (c) a closed path — Conservative work is path independent.

8. Answer: (c) gravity — Gravity is conservative.

9. Answer: (b) friction — Friction depends on path.

10. Answer: (c) work/time — P = W/t.

11. Answer: (a) watt — 1 W = 1 J/s.

12. Answer: (c) 3.6 × 10⁶ J — 1 kWh = 1000 W × 3600 s.

13. Answer: (d) energy — Power × time = energy.

14. Answer: (a) F·v — P = F·v.

15. Answer: (d) ½mv² — K = ½mv².

16. Answer: (d) four times — K ∝ v².

17. Answer: (a) four times — K = p²/2m.

18. Answer: (b) change in kinetic energy — W_net = ΔK.

19. Answer: (a) mgh — Near surface U = mgh.

20. Answer: (d) −GMm/R — Zero at infinity gives negative U.

21. Answer: (d) attractive field — Work must be done to escape.

22. Answer: (c) √(2gR) — v_e = √(2GM/R) = √(2gR).

23. Answer: (c) 11.2 km s⁻¹ — v_e ≈ 11.2 km/s.

24. Answer: (d) mass of escaping body — Mass cancels in ½mv² = GMm/R.

25. Answer: (a) conserved — K + U remains constant.

26. Answer: (c) converted to heat — Total energy is conserved.

27. Answer: (d) potential energy — It can have stored energy.

28. Answer: (c) zero — Gravity is conservative.

29. Answer: (d) elastic potential energy — Area = work stored.

30. Answer: (b) ½kx² — Energy = ½kx².

31. Answer: (b) 720 J — E = Pt = 6 × 120.

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32. Answer: (a) 25 min — t = 3000/2 = 1500 s = 25 min.

33. Answer: (c) 49 kW — P = mgh/t = 1000×9.8×100/20.

34. Answer: (c) 60° — cos θ = 25/(5×10) = 0.5.

35. Answer: (a) independent of mass — W = Fd = K, independent of m.

36. Answer: (a) 4.4 m s^{-1} — v = √(2gΔh) ≈ 4.4 m/s.

37. Answer: (b) gravity — Gravity is conservative.

38. Answer: (d) change in K.E. — $W_{net} = \Delta K$.

39. Answer: (b) joule and kgf·m — Joule is absolute; kgf·m is gravitational.

40. Answer: (a) erg — 1 erg = 1 dyne cm.

41. Answer: (d) −0.28 mgH — K = 0.72 mgH, so friction work = −0.28 mgH.

42. Answer: (d) 600 J — $K \propto v^2$.

43. Answer: (a) 0.75 times — $K \propto mv^2$ gives 3×¼.

44. Answer: (b) kinetic energy — Moving air has K.E.

45. Answer: (a) it has larger mass — K depends on mass as well as speed.

46. Answer: (a) greater K.E. — $K = p^2/2m$.

47. Answer: (b) no net work done — $W_{net} = \Delta K = 0$.

48. Answer: (a) negative work — K.E. decreases.

49. Answer: (a) both K.E. and P.E. — It moves and is in Earth's field.

50. Answer: (b) it depends on path — Longer path gives more heat.

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